

DEREK WONG

Full-Stack Web Developer | Digital Solutions Professional

Singapore | derek.wong.ha@gmail.com | Portfolio | GitHub

PROFESSIONAL SUMMARY

Career-transitioning full-stack web developer with more than 20 years of experience in systems management, technical operations, process improvement and stakeholder coordination. Completed a Professional Diploma in Full Stack Web Development in July 2026 and built a published portfolio of React, Django, MySQL and server-rendered applications, plus a local-AI job-screening automation tool. Combines recent development experience with structured problem-solving, lifecycle planning and technical delivery. Seeking junior, contract, project-based or permanent opportunities in full-stack development, frontend/UI implementation and digital solutions.

TECHNICAL SKILLS

Frontend: HTML, CSS, JavaScript, React, Vite, Bootstrap, responsive UI, semantic templates	Backend: Python, Django, Django REST Framework, REST APIs, JWT authentication
Database & Integration: MySQL, SQL, Django ORM, Google Maps API, Gemini API, HubSpot CRM, Gmail API (OAuth), SQLite	Design & Delivery: Requirements, UML, ERD, wireframes, testing, Git/GitHub, Agile/Scrum
Application Capabilities: Authentication, role-based access, CRUD, search/filtering, forms, uploads	Other Tools: C++, C#, MATLAB, AutoCAD, PSPICE/LTSpice, Microsoft Office, Photoshop, LECTORA

SELECTED DEVELOPMENT PROJECTS

TasteLocal SG - Local Food Tourism Platform

Full-Stack Capstone | React, Django REST Framework, MySQL, JWT

- Delivered multi-role tourist, vendor and administrator workflows for discovery, booking, availability, itinerary planning, reviews and moderation.
- Integrated Google Maps and Gemini recommendations; verified 101 passing backend tests, frontend builds and functional review.

SmartShop AI - AI-Assisted Product Catalogue

Supporting Project | React, Django REST Framework, MySQL, Gemini

- Built a searchable catalogue with 50 controlled products, six categories, synthetic review/purchase data and deterministic AI fallback behaviour.
- Published architecture and documentation with 31 passing backend tests.

HopeHands - Volunteer Onboarding and CRM Workflow

Supporting Project | React, Django REST Framework, MySQL, JWT, HubSpot

- Implemented role-protected onboarding, approval, reporting, CSV intake and failure-aware HubSpot synchronization.
- Added staff-only CRM retry recovery without reversing approval or creating duplicate contacts; verified 56 passing backend tests.

SilentLibrary - Django Library Catalogue

Foundational Coursework | Django, MySQL, Django Templates, HTML, CSS

- Independently built the original server-rendered application without AI assistance; later refined under a human-directed, AI-supported process.
- Implemented searchable catalogue, staff administration, borrowing/return, loan history and custom responsive HTML/CSS; verified 34 passing backend tests.

Career Transition Local AI Assistant - Job Screening Workflow

Personal Automation Project | Python, SQLite, Gmail API, LM Studio (local LLM), openpyxl

- Built a human-in-the-loop workflow that reads job alerts from Gmail (read-only OAuth), removes duplicates, assesses fit with a local LLM grounded in a controlled evidence catalogue, and presents results in a local dashboard for apply/skip decisions.
- Designed for daily use: persistent SQLite state, safe re-runs, retry and recovery handling, and tracker updates only after confirmed submission. Operational since Oct 2026; has screened 250+ listings.

PROFESSIONAL EXPERIENCE

Defence Science and Technology Agency (DSTA)

Engineer / System Manager | Aug 2011 - Feb 2026

- Progressed from engineering support responsibilities to management of high-value technical systems across operational and support lifecycles.
- Managed end-to-end operations and support to maintain availability and meet operational requirements.
- Developed short- and long-term support plans considering mission needs, system condition, lifecycle constraints and resources.
- Managed high-value technical systems, balancing operational availability, lifecycle requirements, risk and stakeholder priorities.
- Reviewed work processes and maintenance approaches to reduce manpower requirements and improve operational efficiency.
- Implemented commercial technologies that reduced costs and procurement lead times.
- Coordinated users, stakeholders, contractors and original equipment manufacturers to resolve technical issues and manage system changes.
- Applied structured problem-solving and data analysis to identify risks, performance trends and improvement opportunities.

Republic of Singapore Navy, Singapore Armed Forces

System Specialist / Technician / Naval Instructor | 2000 - 2011

- Operated, maintained and troubleshoot weapon and electronic systems to support operational readiness.
- Performed preventive and corrective maintenance, fault isolation, calibration, system alignment and technical assessment.
- Supported system installation, testing, commissioning and improvement initiatives.
- Led maintenance and seamanship activities, maintained logistics accountability and deputised the Chief of Department.
- Delivered technical and combat-system training using adult-learning methodologies and served as a subject-matter expert.
- Conducted risk assessments, safety audits and inspections; mentored and assessed junior specialists.

EDUCATION

Lithan Academy Pte. Ltd.

Professional Diploma in Full Stack Web Development | Feb 2026 - Jul 2026

- Intensive full-time programme covering frontend, backend, database, UI and full-stack development, completed through the TasteLocal SG capstone.

National University of Singapore

Bachelor of Technology in Electronics Engineering | Second Upper Honours

Temasek Polytechnic

Diploma in Mechatronics

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Certified Scrum Product Owner | Certified in Cybersecurity (ISC2) | EC-Council Certified Network Defender - Certificate of Attendance | Analytics: From Data to Insights (NUS School of Computing) | Certificate in Methods of Instruction | Adult Learning Methodologies | Presentation and Communication Skills | Dynamics of Leadership | Decision Making and Mind Mapping | Advanced WITS Leader Creativity Tools

SELECTED ACHIEVEMENTS

- DSTA Innovation Award for implementing a solution that reduced cost and procurement lead time.
- Implemented a WITS project achieving approximately S\$300,000 in cost savings.
- Developed a technical-training emulator using LECTORA software.
- Received the Competent Instructor Award and was recognised as Junior Ranking and Petty Officer Vocation Best Trainee.

LANGUAGES

English - Fluent | Mandarin - Fluent